RECTRON
SEMICONDUCTOR
TECHNICAL SPECIFICATION

MD1S
THRU
MD7S

SINGLE-PHASE GLASS PASSIVATED MINI SILICON SURFACE MOUNT BRIDGE RECTIFIER

VOLTAGE RANGE 50 to 1000 Volts CURRENT 0.5 Ampere

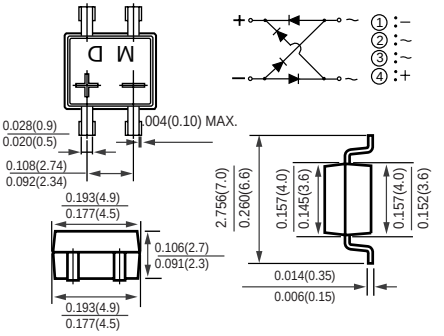
FEATURES

* Surge overload rating - 30 amperes peak
* Ideal for printed circuit board
* Reliable low cost construction utilizing molded
* Glass passivated device
* Polarity symbols molded on body
* Mounting position: Any
* Weight: 0.5 gram

MECHANICAL DATA

* Epoxy : Device has UL flammability classification 94V-0
* UL listed the recognized component directory, file #E94233

MD-S



Dimensions in millimeters

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

Ratings at 25 °C ambient temperature unless otherwise specified.
Single phase, half wave, 60 Hz, resistive or inductive load.
For capacitive load, derate current by 20%.

MAXIMUM RATINGS (At T_A = 25°C unless otherwise noted)

RATINGS	SYMBOL	MD1S	MD2S	MD3S	MD4S	MD5S	MD6S	MD7S	UNITS
Maximum Recurrent Peak Reverse Voltage	V_{RRM}	50	100	200	400	600	800	1000	Volts
Maximum RMS Bridge Input Voltage	V_{RMS}	35	70	140	280	420	560	700	Volts
Maximum DC Blocking Voltage	V_{DC}	50	100	200	400	600	800	1000	Volts
Maximum Average Forward Output Rectified Current at T_A = 30°C - on glass-epoxy P.C.B. (NOTE 1) - on aluminum substrate (NOTE 2)	I_O	0.5 0.8							Amp
Peak Forward Surge Current 8.3 ms single half sine-wave superimposed on rated load (JEDEC method)	I_{FSM}	30							Amps
Operating and Storage Temperature Range	T_J,T_{STG}	-55 to + 150							°C

ELECTRICAL CHARACTERISTICS (At T_A = 25°C unless otherwise noted)

CHARACTERISTICS		SYMBOL	MD1S	MD2S	MD3S	MD4S	MD5S	MD6S	MD7S	UNITS
Maximum Forward Voltage Drop per Bridge Element at 0.5A DC		V_F	1.05							Volts
Maximum Reverse Current at rated DC Blocking Voltage per element	@T_A = 25°C	I_R	10							uAmps
	@T_A = 125°C		0.5							mAmps

NOTE: 1. On glass-epoxy P.C.B. mounted on 0.05 X 0.05" (1.3 X 1.3mm) pads.
2. On aluminum substrate P.C.B. with an area of 0.8 X 0.8 X 0.25" (20 X 20 X 6.4mm) mounted on 0.05 X 0.05" (1.27 X 1.27mm) solder pad.
3. Suffix "-S" Surface Mount for Mini Dip Bridge.

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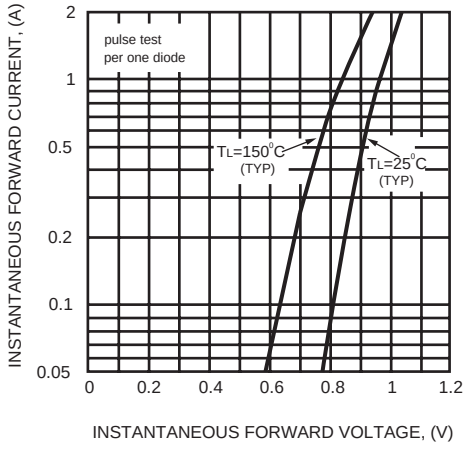
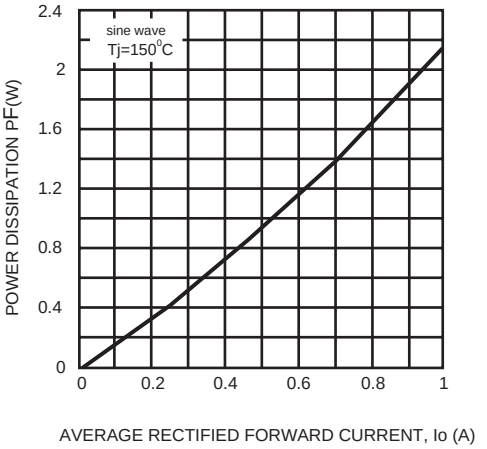
RATING AND CHARACTERISTIC CURVES (MD1S THRU MD7S)

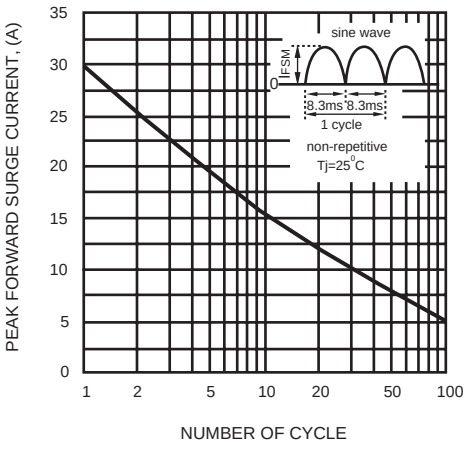
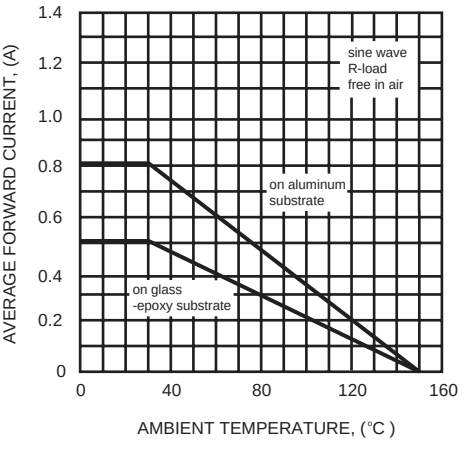
TYPICAL INSTANTANEOUS FORWARD CHARACTERISTICS



POWER DISSIPATION



SURGE FORWARD CURRENT CAPABILITY



TYPICAL FORWARD CURRENT DERATING CURVE